

5.5 Equilibria (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.5 Equilibria (A Level Only)
Difficulty	Easy

Time allowed: 50

Score: /41

Percentage: /100

Question 1a

A saturated solution of silver(I) chloride has a solubility of $1.46 \times 10^{-3} \text{ mol dm}^{-3}$.

Write the expression for the solubility product, K_{sp} , for silver(I) chloride.

[1 mark]

Question 1b

Using the information and your answer given in part (a), calculate the solubility product, K_{sp} , and its units for silver chloride.

Show your working.

$K_{sp} = \dots\dots\dots$

units = $\dots\dots\dots$

[3 marks]

Question 1c

A saturated solution of iron(II) hydroxide has a solubility of $5.82 \times 10^{-6} \text{ mol dm}^{-3}$.

i)

Write the expression for the solubility product, K_{sp} , for iron(II) hydroxide.

[1]

ii)

Calculate the solubility product and units for iron(II) hydroxide. Show your working

$K_{sp} = \dots\dots\dots$

units = $\dots\dots\dots$

[3]

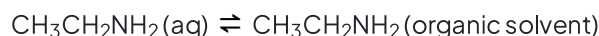
[4 marks]

Question 2a

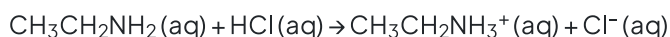
The general equilibrium expression for the partition coefficient is

$$\text{Partition coefficient} = \frac{[\text{equilibrium concentration in mol dm}^{-3} \text{ in organic layer}]}{[\text{equilibrium concentration in mol dm}^{-3} \text{ aqueous layer}]}$$

100 cm³ of a 0.5 mol dm⁻³ solution of aqueous ethylamine, CH₃CH₂NH₂, was shaken with 100 cm³ of an organic solvent at 25 °C and left in a separating funnel to allow an equilibrium to be established. The equilibrium equation is:



Only 50 cm³ of the aqueous layer was run off and titrated against 0.50 mol dm⁻³ dilute hydrochloric acid. 15.0 cm³ of hydrochloric acid was required to reach the end point.



i)

Calculate the amount, in moles, of ethylamine that has reacted with hydrochloric acid.

moles of ethylamine that reacted = mol
[1]

ii)

Calculate the amount, in moles, of ethylamine in the aqueous layer.

moles of ethylamine in aqueous layer = mol
[1]

iii)

Calculate the amount, in moles, of ethylamine in the organic layer.

moles of ethylamine in organic layer = mol
[1]

iv)

Calculate the concentration, in mol dm⁻³, of ethylamine in the aqueous **and** organic layer.

Concentration in aqueous layer = mol dm⁻³Concentration in organic layer = mol dm⁻³
[1]

v)

Calculate the partition coefficient, K_{pc} , of ethylamine between water and the organic solvent. K_{pc} =
[1]

[5 marks]

Question 2b

Use your answer to part (a) to determine whether ethylamine is more soluble in the aqueous or organic solvent. Explain your answer.

Solvent

Explanation

.....

[2 marks]

Question 2c

A solution containing 1.00 g of compound **A** in 100 cm³ of water was shaken with 10 cm³ of ether. 0.60 g of **A** was transferred to the ether layer.

Calculate the partition coefficient of compound **A** between ether and water. Show your working.

$$K_{pc} = \dots\dots\dots$$

[3 marks]**Question 3a**

The K_{sp} value for silver bromide, AgBr(s) is 3.3×10^{-13} at 298 K.

Write the equilibrium equation for the changes that happen in a saturated solution of silver bromide in the presence of some solid.

[1 mark]**Question 3b**

A solution of sodium bromide was added to a saturated solution of silver bromide solution. How would the concentration of dissolved silver ions change. Explain your answer.

Change in concentration of silver ions

Explanation

.....

[2 marks]

Question 3c

Calculate the concentration, in mol dm^{-3} , of silver ions at 298 K in a saturated solution of silver bromide. Show your working.

$[\text{Ag}^+(\text{aq})] = \dots\dots\dots \text{mol dm}^{-3}$

[2 marks]

Question 4a

Acidic buffer solutions are made of a weak acid and its salt. pH calculations of buffer solutions can be conducted using the same equation used for weak acids:

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

State the assumptions made when using this equation for buffer solutions.

[3 marks]

Question 4b

Suggest a salt that could be used to form a buffer solution with propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$.

[1 mark]

Question 4c

A buffer solution contains a 0.15 mol dm^{-3} concentration of propanoic acid and a propanoate salt with a concentration of $[A^-] = 0.30 \text{ mol dm}^{-3}$.

At 298 K the value of K_a for propanoic acid is $1.35 \times 10^{-5} \text{ mol dm}^{-3}$.

i)

Calculate the concentration of $[H^+]$ in this buffer solution at 298 K. Give your answer to 2 decimal places. Show your working.

$[H^+] = \dots\dots\dots \text{mol dm}^{-3}$
[3]

ii)

Calculate the pH of the buffer solution to 2 decimal places.

pH = $\dots\dots\dots$
[1]

[4 marks]

Question 4d

Buffer solutions can be tailored to a specific pH by 'coarse-tuning' by changing the acid used, and 'fine-tuning' by altering the ratio of $[HA] : [A^-]$.

Predict and explain the effect on the pH of the buffer solution by increasing the concentration of the salt only.

[2 marks]

Question 5a

Two acids used in the same laboratory, which both have the same concentration, 0.1 mol dm^{-3} , can have significantly different pH values. Explain why.

[2 marks]

Question 5b

Calculate the pH of a solution of 0.1 mol dm^{-3} HCl.
Show your working

pH =

[2 marks]

Question 5c

A sample of 0.1 mol dm^{-3} methanoic acid has a K_a value of $1.78 \times 10^{-4} \text{ mol dm}^{-3}$ at 298 K.

i)

Write an expression for the acid dissociation constant, K_a , for methanoic acid.

$K_a =$

[1]

ii)

Calculate the pH of the 0.1 mol dm^{-3} methanoic acid. Give your answer to **two** decimal places and show your working.

pH =

[3]

[4 marks]

Model Answers: Easy

1a

a) The expression for the solubility product, K_{sp} , for silver(I) chloride is:

- $K_{sp} = [Ag^+(aq)][Cl^-(aq)]$; [1 mark]

[Total: 1 mark]

- When writing the expression for K_{sp} you need to
 - Break up the ionic compound into its ions
 - Write in the charges of the ions
 - Write in (aq)
 - Make sure you use square brackets
 - Raise to the power of their relative concentrations
 - e.g. If you have $MgCl_2$, the K_{sp} expression is $K_{sp} = [Mg^{2+}(aq)][Cl^-(aq)]^2$

1b

b) To calculate the solubility product, K_{sp} , and units of silver chloride:

- $K_{sp} = [1.46 \times 10^{-3}][1.46 \times 10^{-3}]$; [1 mark]
- $K_{sp} = 2.13 \times 10^{-6}$; [1 mark]
- Units = $mol^2 dm^{-6}$; [1 mark]

[Total: 3 marks]

- To set out this calculation you can:
- Write the solubility product
 - $K_{sp} = [Ag^+(aq)][Cl^-(aq)]$
- Write in the values
 - $K_{sp} = [1.46 \times 10^{-3}][1.46 \times 10^{-3}]$
 - $K_{sp} = 2.13 \times 10^{-6}$
- To calculate the units
 - Write in $mol dm^{-3}$ for each value

- $\text{mol dm}^{-3} \times \text{mol dm}^{-3}$
- Add the indices together
- $\text{mol}^2 \text{dm}^{-6}$

1c

c)

i) The expression for the solubility product, K_{sp} , for iron(II) hydroxide is:

- $K_{\text{sp}} = [\text{Fe}^{2+}(\text{aq})][\text{OH}^{-}(\text{aq})]^2$; [1 mark]

ii) To calculate the solubility product and units for iron(II) hydroxide:

- $K_{\text{sp}} = [5.82 \times 10^{-6}][2 \times 5.82 \times 10^{-6}]^2$; [1 mark]
- $K_{\text{sp}} = 7.89 \times 10^{-16}$; [1 mark]
- Units = $\text{mol}^3 \text{dm}^{-9}$; [1 mark]

[Total: 4 marks]

- The equation for the formation of the ions is
 - $\text{Fe}(\text{OH})_2(\text{s}) \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq})$
- For this calculation, you need to double the concentration of the hydroxide ions as well as raising to the power of their relative concentrations
- Again, to calculate the units write mol dm^{-3} for each value
- In this case you need three mol dm^{-3} and then add the indices together
 - $\text{mol dm}^{-3} \times \text{mol dm}^{-3} \times \text{mol dm}^{-3} = \text{mol}^3 \text{dm}^{-9}$

2a

a)

i) To calculate the amount, in moles, of ethylamine has reacted with hydrochloric acid:

- $n(\text{HCl}) = n(\text{CH}_3\text{CH}_2\text{NH}_2) = 0.5 \times 0.015 = 0.0075 \text{ (mol)}$; [1 mark]

ii) To calculate the amount, in moles, of ethylamine in the aqueous layer:

- $2 \times 0.0075 = 0.015$; [1 mark]

iii) To calculate the amount, in moles, of ethylamine in the organic layer:

- Total $n(\text{CH}_3\text{CH}_2\text{NH}_2) = 0.5 \times 0.10 = 0.05 \text{ (mol)}$

AND

$$n(\text{CH}_3\text{CH}_2\text{NH}_2) \text{ (organic layer)} = 0.05 - 0.015 = 0.035; \text{ [1 mark]}$$

iv) To calculate the concentration, in mol dm^{-3} , of ethylamine in the aqueous **and** organic layer:

- $[\text{CH}_3\text{CH}_2\text{NH}_2 \text{ (aq)}] = 0.15 \text{ (mol dm}^{-3}\text{)}$

AND

$$[\text{CH}_3\text{CH}_2\text{NH}_2 \text{ (organic layer)}] = 0.35 \text{ (mol dm}^{-3}\text{)}; \text{ [1 mark]}$$

v) To calculate the partition coefficient, K_{pc} , of ethylamine between water and the organic solvent:

- $K_{\text{pc}} = \frac{0.35}{0.15} = 2.3; \text{ [1 mark]}$

[Total: 5 marks]

- This question has been broken down step by step
- Take your time to read the question
 - Many students will miss the fact that only 50 cm^3 of the aqueous layer has been used in the titration, so the overall moles of the ethylamine solution is double the number of moles calculated in the titration

2b

b) Ethylamine is more soluble in:

- Organic solvent; [1 mark]
- Value for K_{pc} is above 1; [1 mark]

[Total: 2 marks]

- If the value of K_{pc} is larger than 1 the solvent is more soluble in the organic solvent than in water

2c

c) To calculate the partition coefficient:

- $K_{sp} = \frac{[\text{compound A (ether)}]}{[\text{compound A (aq)}]}$; [1 mark]
- $[\text{compound A (ether)}] = \frac{0.60}{10} \times 1000 = 60 \text{ g dm}^{-3}$

AND

- $[\text{compound A (aq)}] = \frac{0.40}{100} \times 1000 = 4 \text{ g dm}^{-3}$; [1 mark]
- $K_{sp} = \frac{60}{4} = 15$; [1 mark]

[Total: 3 marks]

- **Careful:** You are told that 0.60 g of compound A is extracted from 10 cm³ so to calculate [compound A] you must divide by 10 cm³ and not 100 cm³

3a

a) The equilibrium equation for the changes that happen in a saturated solution of silver bromide in the presence of some solid is:

- $\text{AgBr(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq})$; [1 mark]

[Total: 1 mark]

- 1 mole of silver chloride will dissolve to give 1 mole of silver ions and 1 mole of chloride ions

3b

b) The concentration of silver ions will change by:

- Concentration of silver ions / Ag⁺ ions will decrease; [1 mark]
- Adding more bromide ions will shift equilibrium to the left hand side / LHS; [1 mark]

[Total: 2 marks]

- The equation is:

[Total: 3 marks]

- **Careful:** You are told that 0.60 g of compound **A** is extracted from 10 cm³ so to calculate [compound **A**] you must divide by 10 cm³ and not 100 cm³

3a

a) The equilibrium equation for the changes that happen in a saturated solution of silver bromide in the presence of some solid is:

- $\text{AgBr(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq})$; [1 mark]

[Total: 1 mark]

- 1 mole of silver chloride will dissolve to give 1 mole of silver ions and 1 mole of chloride ions

3b

b) The concentration of silver ions will change by:

- Concentration of silver ions / Ag^+ ions will decrease; [1 mark]
- Adding more bromide ions will shift equilibrium to the left hand side / LHS; [1 mark]

[Total: 2 marks]

- The equation is:
 - $\text{AgBr(s)} \rightleftharpoons \text{Ag}^+(\text{aq}) + \text{Br}^-(\text{aq})$
- Adding more bromide ions will increase the concentration of bromide ions
- According to Le Chatelier's principle, the increase in concentration will cause the equilibrium to shift to the side with the lowest concentration of bromide ions (left) to oppose the change

3c

c) To calculate the concentration of silver ions:

- $[\text{Ag}^+(\text{aq})][\text{Br}^-(\text{aq})] = 3.3 \times 10^{-13}$; [1 mark]
- $[\text{Ag}^+(\text{aq})] = \sqrt{3.3 \times 10^{-13}} = 5.74 \times 10^{-7} (\text{mol dm}^{-3})$; [1 mark]

[Total: 2 marks]

- The key is to write the correct K_{sp} expression for AgBr which is
 - $K_{sp} = [Ag^+(aq)][Br^-(aq)] = 3.3 \times 10^{-13}$
- The K_{sp} value is given in part (a)

4a

a) The assumptions needed when using this equation for buffer solutions are:

- $[HA]$ = the concentration of the weak acid used; [1 mark]
- $[A^-]$ = the concentration of ($[A^-]$ in) the salt of the acid being used; [1 mark]
- $[H^+]$ is negligible/ too small to account for/ minimal dissociation of the acid occurs; [1 mark]

[Total: 3 marks]

- In order to correctly use the equation for buffer solutions it is important to know the assumptions being made
- For most salts used the value of $[A^-]$ will equal the concentration of the salt as long as there is only 1 mole of A^- in the formula

4b

b) A salt that could be used to form a buffer solution with propanoic acid is:

- Sodium propanoate/ CH_3CH_2COONa
OR
Potassium propanoate/ CH_3CH_2COOK ; [1 mark]

Allow other soluble salts that include the $CH_3CH_2COO^-$ ion

[Total: 1 mark]

- You would also gain the mark for giving other soluble salts that include the $CH_3CH_2COO^-$ ion
- A weak acid can form a buffer solution with the salt of the weak acid; Group 1 metals always form soluble salts so sodium and potassium salts are the standard ones used, but lithium salts would work just as well

4c

c)

i) The concentration of $[H^+]$ in the buffer solution is:

- $K_a = \frac{[H^+][A^-]}{[HA]}$; [1 mark]
- $[H^+] = \frac{K_a \times [HA]}{[A^-]}$; [1 mark]
- $[H^+] = \frac{(1.35 \times 10^{-5} \times 0.15)}{0.30} = 6.75 \times 10^{-6}$; [1 mark]

ii) The pH of the buffer solution is:

- $pH = -\log(6.75 \times 10^{-6}) = 5.17$; [1 mark]

[Total: 4 marks]

- The equation must be rearranged to find $[H^+]$
- In this reaction:
 - The concentration of $[A^-]$ is given in the question
 - The concentration of $[HA]$ is the same as the concentration of the propanoic acid given in the question

4d

d) The effect on the pH of the buffer when the concentration of the salt is increased is:

- The pH will increase; [1 mark]
- As $[H^+]$ will decrease; [1 mark]

[Total: 2 marks]

- The equation $[H^+] = \frac{K_a \times [HA]}{[A^-]}$ shows that as the value of $[A^-]$ increases the value of $[H^+]$ will decrease as the size of the fraction will decrease
- A lower $[H^+]$ means a higher pH

5a

a) Two acids in the same conditions with the same concentration can have different pH values as:

- Weak acids have different strengths/ the acids could have different K_a values/ one acid could be a strong acid and the other be a weak acid; [1 mark]
- pH depends only on the extent of the dissociation of H^+ ions/ the concentration of H^+ in the solution/ strong acids fully dissociate and weak acids only partially dissociate; [1 mark]

[Total: 2 marks]

- It seems counterintuitive that solutions with equal concentrations on a bottle have different pH values, but this is about understanding what makes a solution an acid and why acids react differently
- The pH is based on the concentration of H^+ , which in turn depends on how much that acid dissociates, or ionises
- Strong acids ionise fully, so the concentration of H^+ is equivalent to the concentration of HA for monobasic acids
- Weak acids partially dissociate, so the assumption is that there is minimal dissociation and most HA remains as HA

5b

b) The pH of the solution can be calculated by:

- $[H^+] = 0.1 \text{ mol dm}^{-3}$; [1 mark]
- $pH = -\log_{10}[H^+] = -\log_{10}(0.1) = 1.0$; [1 mark]

[Total: 2 marks]

- As strong acids fully dissociate, the $[H^+]$ is equal to the concentration of the acid
- $HA \rightleftharpoons H^+ + A^-$
- Therefore in the 0.1 mol dm^{-3} solution, the $[H^+]$ is also 0.1 mol dm^{-3}
- This value is then put straight into the pH equation to calculate pH

- As strong acids fully dissociate, the $[H^+]$ is equal to the concentration of the acid
- $HA \rightleftharpoons H^+ + A^-$
- Therefore in the 0.1 mol dm^{-3} solution, the $[H^+]$ is also 0.1 mol dm^{-3}
- This value is then put straight into the pH equation to calculate pH

5c

c)

i) The K_a expression for methanoic acid is:

- $K_a = \frac{[H^+][HCOO^-]}{[HCOOH]}$; [1 mark]

ii) The pH of 0.1 mol dm^{-3} methanoic acid is calculated as:

- $[H^+]^2 = K_a \times [HCOOH]$
OR
 $[H^+]^2 = 1.78 \times 10^{-4} \times 0.1 = 1.78 \times 10^{-5}$; [1 mark]
- $[H^+] = \sqrt{1.78 \times 10^{-5}}$
AND
 $[H^+] = 4.22 \times 10^{-3} \text{ mol dm}^{-3}$; [1 mark]
- $\text{pH} = -\log_{10}[H^+] = 2.37$; [1 mark]

[Total: 4 marks]

- The expression for K_a must involve methanoic acid, not just the general expression for K_a to achieve the mark
- In the calculation, the first mark is to correctly rearrange the expression to find $[H^+]^2$
- The second and third marks are for giving the correct values for $[H^+]^2$ and $[H^+]$
- The final mark is for the pH correctly calculated **AND** given to two decimal places

